

PMOM Neutralization — planning document

Suggested document structure first, then detailed explanations to draw from when writing each section

This document has two parts. **Part A** is a recommended structure for the final write-up you will send to Matt. **Part B** contains the substantive analysis and explanations that feed into each section — the material we worked through in conversation, organized so you can lift it into the right places. Write Part A in order; pull from Part B as needed.

Part A — Recommended document structure

Section 0 — Abstract / Executive summary

Four short paragraphs at the top of the document, before the methodology:

- **What we tested.** PMOM-neutralized V and Q scores at three levels: factor score, standalone L/S, and WFP composite (in three variants: P / VP / EVP).
- **What we found at the score and standalone level.** The theory works. EV is the most PMOM-contaminated factor; neutralization cleans it cleanly. The DD-regime flip predicted by the theory shows up clearly for V and EV, weakly or not at all for the other factors.
- **What we found at the WFP composite level.** WFP_P (full neutralization) reverses the regime hedge sharply: it performs *worse* in PMOM-tailwind regimes (Pearson -0.91 cross-universe). WFP_EVP (EV-only) moves directionally in line with theory ($+0.59$) but with a magnitude too small to register at the IR level (full-sample Δ IR -0.02).
- **What this means for implementation.** The theory is correct; the composite architecture is the implementation problem. WFP_EVP is the minimal-disruption operating point. Capturing the full hedge at the composite level would require rebuilding the WFP from independent legs rather than swapping cleaned inputs into the existing recipe.

Section 1 — Methodology

Keep the structure you have. Order the content as:

- B-track vs P-track definition (Baseline scores vs PMOM-neutralized scores).
- WFP variants: Baseline / P / VP / EVP, with explicit weights of what each variant changes.
- Active-weight construction (your existing formulation).
- DD conditioning (low DD = $\text{DDSmooth} \geq 0$, high DD = $\text{DDSmooth} < 0$).
- **The Value/PMOM identity** as a short sub-section — this is the theoretical anchor for the whole analysis. Lift directly from your ANALYSIS_NOTES.md. State that the identity predicts every Value factor of the form F/M carries an anti-PMOM tilt by construction.

Section 2 — Validation of the neutralization

- **2.1 Cross-sectional score correlations** (your existing 1.2): show neutralization produces XS score correlations with PMOM close to zero across all factors, with reduced std deviation over time. One paragraph: "The cleanup works at the score level."
- **2.2 L/S return correlations with PMOM** (your existing 1.1): show the residual anti-correlation at the return level after neutralization. EV stays the most negative even after cleanup, consistent with the residual term in the Value/PMOM identity having low noise relative to the $-PMOM$ term. Other factors clean up to near zero.

Section 3 — Standalone L/S performance

Your existing Section 2. Lead with the cross-universe table. Reading: modest IR improvement for V (+0.03) and EV (+0.06); modest IR deterioration for Q (−0.07). Magnitudes are small enough that this section alone does not make the case for adoption.

Section 4 — DD regime conditioning (the factor-level theory test)

This is your theory confirmation. Reframe it to lead with the EV regime flip, not the table. The narrative line for this section is in Part B below.

- Lead paragraph: state the theory's prediction and the EV result (Δ IR +0.10 in low DD, −0.07 in high DD).
- Use the absence of a flip in Q / Profit / Investment / FCF_Value / TYTD2EV1Y as *supporting* evidence — they don't flip because they don't have the contamination to remove.
- Cross-universe table comes second, supporting the lead.
- Note the cross-universe consistency: SPX (biggest contamination) shows the biggest flip; MXAP/MXEF (smaller contamination) show smaller but directionally consistent flips.

Section 5 — WFP composite performance (the implementation test)

Three sub-sections, written in order:

- **5.1 Full-sample IR comparison** (your existing Section 4 table). Lead: "All three neutralized variants underperform Baseline. WFP_P is the worst (Δ IR −0.04), EVP is the smallest move (Δ IR −0.02)."
- **5.2 Regime-conditional behavior** — the tercile-by-PMOM table from your findings_evp_vs_pmom.md (WFP_P delta of −34.6 bps in M_HIGH). This is the smoking gun. Lead with the cross-universe Pearson summary (WFP_P at −0.91, EVP at +0.59).
- **5.3 Why the composite reverses** — the V/M overlap mechanism. Detailed narrative below in Part B. This sub-section explains both why WFP_P breaks *and* why WFP_EVP delivers only a small directional effect.

Section 6 — Annual returns confirmation

Your existing Section 5. Show one cross-universe summary table of (Pearson, Spearman, Pearson-monthly, SignMatch %) for WFP_X – WFP_B vs PMOM_3SLOW. Then individual universe annual tables as appendix-style backing evidence. This section is confirmation of Section 5.2 over longer windows, no new arguments.

Section 7 — Conclusions and open questions

- **Conclusion 1.** Matt's hypothesis is correct at the factor score and standalone level. The Value/PMOM identity is empirically confirmed by Section 4 (DD regime flip for EV).
- **Conclusion 2.** Naive implementation (WFP_P) breaks the composite's implicit V/M overlap risk reduction. This is not noise — it's mechanical and shows up consistently across universes (Pearson −0.91 cross-universe).

- **Conclusion 3.** EVP is the minimal-disruption variant; the regime hedge appears at the predicted magnitude (small, ~12 bps in 2024 SPX) without breaking the composite.
- **Open question 1.** Re-architect WFP from clean V + clean Q + standalone M as independent legs, designed without relying on V/M overlap for risk reduction. Does this unlock the regime hedge at a meaningful magnitude?
- **Open question 2.** DD-conditional EVP (low DD → EVP scores, high DD → Baseline scores). Does it dominate both endpoints in IR?
- **Sanity-check addition.** Show 12M MOM neutralization results alongside PMOM neutralization to demonstrate the choice of momentum target doesn't materially change the conclusions (since PMOM is largely built from 12M MOM components).

Appendix — per-universe tables

All per-universe tables (Section 1, 2, 3 by universe; Section 5 annual returns by universe) go here. Cross-universe means stay in the main body; per-universe detail is reference material. This keeps the main narrative readable.

Part B — Substantive explanations to draw from

What follows is the analytical material we worked through. Each block maps to a section in Part A. Lift, paraphrase, or compress as needed when writing.

B.1 — DD regime conditioning (for Section 4)

The theory prediction, stated cleanly. In low DD (PMOM works → V has a headwind from its negative correlation with PMOM), neutralizing V should remove the drag, so PMOM-neutralized V should outperform Baseline V. In high DD (PMOM crashes → V's anti-PMOM tilt becomes a tailwind), neutralizing V removes that natural hedge, so PMOM-neutralized V should underperform Baseline V. The clean signature of the theory is a **directional flip** in the Delta IR between regimes: positive in low DD, negative in high DD.

What your cross-universe Section 3 table shows. For Value and especially Earnings_Value, the regime flip is exactly as predicted:

Factor	Low DD Δ IR	High DD Δ IR	Verdict
Value	+0.06	−0.03	Flip — theory ✓
Earnings_Value	+0.10	−0.07	Strong flip — theory ✓
FCF_Value	−0.05	+0.06	No flip
TYTD2EV1Y	−0.07	0.00	No flip
Quality	−0.11	+0.01	No flip
Profit	−0.07	+0.01	No flip
Investment	−0.05	+0.01	No flip

How to read this. EV's Delta IR swings from +0.10 in low DD to −0.07 in high DD — a 0.17 IR swing, the largest of any factor and the cleanest possible confirmation of the theory. For factors without a structural anti-PMOM tilt (Quality, Profit, Investment, FCF_Value, TYTD2EV1Y), neutralization either hurts in both regimes or shows no consistent pattern. This is consistent with the theory: removing a contamination that isn't really there adds noise rather than signal. **The regime flip is concentrated where the Value/PMOM identity predicts it should be.**

Cross-universe consistency check. The strongest EV flip is in SPX (low DD +0.16 IR / high DD −0.20 IR — a 0.36 swing). This matches Section 1: SPX is the universe where the L/S return anti-correlation between Earnings_Value and PMOM is strongest (−0.37 baseline). Bigger drag to remove → bigger regime effect when you remove it. MXEF and MXAP show smaller flips in the right direction, consistent with their smaller baseline contamination. This is an additional consistency check the theory passes.

Suggested section 4 lead paragraph:

Section 4 — Performance by DD regime confirms the theory at the EV level. The Value/PMOM identity predicts that neutralizing the anti-PMOM tilt should help when PMOM is rewarded (low DD) and hurt when PMOM is losing (high DD). For Earnings_Value, the cross-universe mean Delta IR goes from +0.10 in low DD to -0.07 in high DD — exactly the regime flip the theory predicts, and the largest swing of any factor. Value as a composite shows a smaller but directionally identical flip (+0.06 → -0.03). For factors without a structural anti-PMOM tilt — Quality, Profit, Investment, FCF_Value, TYTD2EV1Y — neutralization either hurts in both regimes or shows no consistent pattern. This is consistent with the theory: removing a contamination that isn't really there adds noise rather than signal. The regime flip is concentrated where the identity predicts it should be.

B.2 — Why WFP_EVP doesn't meaningfully outperform (for Section 5.3)

The puzzle, stated precisely. At the EV component level, neutralizing EV to PMOM gives a clean regime flip (Δ IR +0.10 in low DD, -0.07 in high DD), and the 2024 SPX delta is +0.60 pp when PMOM did +11.2%. At the WFP_EVP level — where EV is the only thing changed — the cross-universe full-sample IR is 1.43 vs Baseline 1.45 (Δ -0.02), and the regime correlation is +0.59 directionally, but the magnitude is too small to register at the IR level. **Why does the component-level benefit not translate to the composite?**

There are three things stacked on top of each other, and you have to separate them.

1. The score change is small by construction. WFP_EVP has score correlation 0.999 with Baseline. That's not a typo. EV is one of three Value sub-themes (37.5% weight), and Value is one of three legs in the WFP (33% weight). So EV's effective weight in WFP is roughly $0.375 \times 0.333 \approx 12.5\%$. Neutralizing 12.5% of the WFP score against PMOM is a near-no-op at the composite score level. This is the first piece of the answer: even if the EV neutralization were perfectly effective at the EV level, its share of the WFP is small enough that the composite-level impact would be small.

2. The V/M structural overlap absorbs much of what's left. The WFP holds active weights from V, Q, M legs. In SPX, V has return correlation -0.29 with M and EV has -0.37. That means V and M are taking opposite-signed positions on overlapping stocks — when M says long Stock A (winner), V often says short Stock A (price ran up → expensive). These positions cancel partially at the WFP active-weight level. The cancellation is a free risk reduction: WFP_B has lower vol than the unweighted sum of V + Q + M would have. When you neutralize EV, you're removing some of that opposition specifically in the EV-driven positions. WFP_EVP active weights on those stocks are now more aligned with M's bet — the cancellation that was reducing risk is partially undone.

The data confirms this directly. WFP_EVP vol is 5.22% vs Baseline 5.16% — tiny (6 bps) but in the predicted direction. WFP_VP is 5.30% (more vol added). WFP_P is 4.22% — here the mechanism is different and is what blocks the regime hedge entirely (covered separately in the WFP_P section). For EVP, the vol increase is real but small because the position change is small.

3. The benefit you'd expect from neutralizing EV is also small at the composite level for the same reason. EV is 12.5% of WFP. Even if PMOM-neutralizing EV gives EV +5 bps per +1% of M return (which your regression shows it does), the WFP only gets 12.5% of that — about 0.6 bps per +1% of M. Over a year where M does +10%, that's ~6 bps to the WFP return. Real, in the right direction, but indistinguishable from noise at the IR level.

The composite-level result for WFP_EVP is exactly what the math predicts. A small positive directional effect (regime correlation +0.59, sign-match 70%), dominated by the share of EV in the WFP. The theory is not failing at the composite — it's operating at a scale where the effect is hard to see.

Suggested Section 5.3 narrative for WFP_EVP:

The composite-level result for WFP_EVP is exactly what the math predicts: a small positive directional effect, dominated by the share of EV in the WFP. EV represents roughly 12.5% of

the WFP score (37.5% within Value $\times$ 33% within the WFP recipe). PMOM-neutralizing EV changes the composite score by a near-no-op (score correlation 0.999 with Baseline). The directional effect predicted by theory still appears: WFP_EVP returns are positively correlated with PMOM at +0.59 cross-universe, sign-match 70%. But the magnitude scales with EV's share of the composite, so the effect is small (~12 bps in 2024 SPX, full-sample Δ IR -0.02). At the same time, removing EV's anti-PMOM tilt also removes a small piece of the V/M structural cancellation that was contributing to Baseline WFP's vol reduction. WFP_EVP vol is 0.06% higher than Baseline — small, in the predicted direction, and a hint of the larger mechanism that drives the much worse WFP_P result.

B.3 — Why WFP_P fails (for Section 5.3, continued)

WFP_P is not the same story as WFP_EVP at higher magnitude — it's a **different mechanism**, or rather the same mechanism scaled up to where it dominates.

When V and Q are neutralized:

- V_clean is no longer anti-M → the V/M cancellation that gave Baseline its diversification is gone.
- Q_clean is now slightly anti-M (it was mildly pro-M before) → it actively subtracts return when M wins.
- The composite becomes structurally concentrated on momentum: vol drops sharply (4.22% vs 5.16% Baseline) but return drops faster (6.02% vs 7.54%).
- The regime correlation with PMOM flips strongly negative (Pearson -0.91 cross-universe).

The fix works against the WFP's existing risk architecture. The Baseline WFP was implicitly using V's anti-M tilt as a hedge. Removing the tilt removes the hedge. Whether this is a feature or a bug depends on whether you trust the structural overlap to keep producing the right kind of risk reduction in the future, or whether you want clean independent factor bets and accept higher composite vol in exchange.

Suggested Section 5.3 narrative for WFP_P:

The puzzle of WFP_P is the same mechanism as WFP_EVP, amplified. When V and Q are both neutralized, the position changes are no longer minor. V_clean is no longer anti-M, so the V/M cancellation that gave Baseline its diversification is gone. Q_clean is now slightly anti-M (it was mildly pro-M before), so it actively subtracts return when M wins. The composite becomes structurally concentrated on momentum: vol drops (4.22% vs 5.16% Baseline) but return drops faster (6.02% vs 7.54%), and the regime correlation with PMOM flips strongly negative (Pearson -0.91 cross-universe). The fix works against the WFP's existing risk architecture, which was implicitly using V's anti-M tilt as a hedge.

B.4 — What this means for implementation (for Section 7)

The component theory is correct; the WFP composite architecture is built on top of mechanisms that the theory's fix would partially undo. WFP_EVP is the operating point where the size of the intervention is small enough that the regime hedge appears in the right direction without breaking the composite, at the cost of small magnitude. To capture the full theoretical benefit at the WFP level would require **rebuilding** the WFP from independent V_clean, Q_clean, M legs — explicitly designed without relying on V/M overlap for risk reduction. That's the open question for follow-up work.

The three-sentence summary, for the meeting:

WFP_EVP isn't broken. It's working exactly as the theory predicts at the magnitude the theory predicts. The magnitude is just too small to see clearly at the composite IR level, because EV is only ~12.5% of the WFP. To get a meaningful composite-level effect, we'd need to either neutralize more (which WFP_P shows breaks the V/M overlap risk architecture) or rebuild the

WFP without the overlap dependency.

B.5 — The 12M MOM neutralization sanity check (for Section 7)

You have 12M MOM neutralization results but they aren't in the current document. Worth adding as a short sanity-check note. PMOM is constructed from 12M MOM, 11M MOM, and 12M MOM exhaustion components, so neutralizing to overall 12M MOM should produce results very close to neutralizing to PMOM. The value of showing it: it lets you say “we tested two natural neutralization targets and both tell the same story.” If the patterns are nearly identical, mention it in one paragraph and move on. If they diverge meaningfully — for example, if 12M MOM neutralization preserves more of the V/M overlap and consequently shows a smaller WFP-level reversal — that's important on its own and reshapes some of the conclusions.